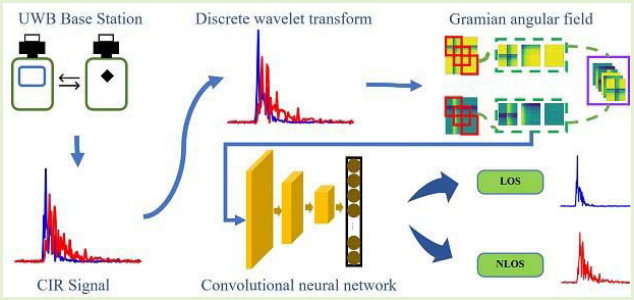
UWB NLOS Recognition Based on Improved Convolutional Neural Network Assisted by Wavelet Analysis and Gramian Angular Field

Xiaogang Deng, *Member, IEEE*, Zhiyuan Ping, and Rui Sun

***Abstract*—In the ultrawideband (UWB)-based indoor wireless positioning scene, nonline-of-sight (NLOS) is a common phenomenon due to the existence of obstacles and it may result in significant positioning errors. So, it is an important research issue to recognize the UWB NLOS signals from the line-of-sight (LOS) signals. As an emerging deep-learning technology, a convolutional neural network (CNN) has been applied successfully to handle this issue. However, the traditional CNN omits the influence of measurement noises and does not consider the mining of the signal's temporal relationship. To solve this problem, a UWB NLOS recognition method, called wavelet Gramian CNN (WGCNN), is proposed by integrating the wavelet analysis and the Gramian angular field (GAF). First, the discrete wavelet transform (DWT) is used to extract the trend curve of the channel impulse response (CIR) signals by eliminating the measurement noises. Furthermore, two GAFs, consisting of Gramian angular summation field (GASF) and Gramian angular difference field (GADF), are introduced to transform the 1-D CIR signal into the 2-D images, which describe the temporal correlation of the signals. Moreover, a GAF image reorganization strategy is designed to capture the key subimages to eliminate the influence of invalid image parts. Finally, the reorganized subimages are fed into the CNN model to build an LOS/NLOS classifier. The experiments are performed on a benchmark dataset provided by the EU "Horizon 2020" program, and the results show that the proposed WGCNN method has higher recognition accuracy than the traditional CNN method.**



***Index Terms*— Convolutional neural network (CNN), discrete wavelet transform (DWT), Gramian angular field (GAF), nonline-of-sight (NLOS).**

I. Introduction

With the development of science and technology, wireless positioning systems are penetrating industrial production and social life [1], [2], [3]. The common positioning scenes can be categorized into two classes: outdoor and indoor. The outdoor positioning technologies, usually based on the satellites or the base stations [4], are relatively mature. By contrast, indoor positioning technology is more challenging due to the complex indoor environment and the high-performance requirement. Some typical indoor positioning methods include ultrawideband (UWB), Bluetooth, Wi-Fi, and ultrasonic [5], [6]. Among them, UWB has received remarkable focus due to its advantages of high positioning accuracy, large system capacity, and low power consumption [7]. Nevertheless, different kinds of obstacles often exist in the indoor environment. This leads to the occurrence of nonline-of-sight (NLOS) signals and has a great impact on the accuracy of indoor positioning [8], [9]. Therefore, the NLOS signal recognition [10], [11] has become an important research topic in the UWB-based indoor positioning field.

The existing UWB NLOS signal recognition methods include distance-based methods, position-based methods, and channel-based methods [12]. The first two kinds of methods require some context information, such as distance or position, while the last kind of method only relies on the UWB channel signal, referred to as the channel impulse response (CIR). Guo et al. [13] proposed a position-based method, which computes the residual between the initial and improved position estimator and builds a hypothesis testing algorithm to identify the NLOS signals. Wu et al. [14] discussed a distance-based method, which investigates the correlation of the measured TDOA distances and applies a

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distance-based classifier to discriminate the NLOS signals. The CIR waveform is obviously affected by the transmission path, and different propagation paths have different impulse responses. Therefore, the CIR characteristic parameters can be extracted from the CIR waveform for NLOS identification [15]. The channel-based methods are more efficient than the other two kinds of methods because they directly analyze the original channel information without the requirement of complex distance and position computations.

In recent years, channel-based NLOS recognition has attracted wide attention from researchers, which is intrinsically a classification task between line-of-sight (LOS) and NLOS signals. The present methods generally adopt two processing ways. One is to first extract the features of CIR signals and then build an LOS/NLOS classifier with some classical machine-learning tools. Yu et al. [16] extracted eight CIR features, including the total energy, the maximal amplitude, and the normalized strongest path energy, and applied the fuzzy comprehensive evaluation (FCE) to identify the NLOS signal. Yang et al. [17] used six features as the input of the import vector machine (IVM) for NLOS identification. In Sang et al.'s work [18], a total of 12 features are constructed and different machine-learning tools, such as support vector machines (SVMs), random forest (RF), and multilayer perceptron, are applied to build the LOS/NLOS classifier. To sum up, in this kind of method, the key point is the construction of features, which are highly dependent on the user experience. To avoid the severe dependence on feature engineering, the other way is put forward by performing the deep-learning modeling directly on the CIR signals without the manual feature extraction [19]. The commonly used deep-learning model in this field is the convolutional neural network (CNN). Bregar and Mohorcic [20] proposed an NLOS channel classification method by using the CNN to process the raw CIR data. Considering the possible information loss, Cui et al. [21] applied an improved CNN model, called the capsule network, to deal with the LOS/NLOS classification issue. Furthermore, Jiang et al. [19] directly combined CNN and LSTM neural networks to deal with the CIR signal sequence. Considering the NLOS classification accuracy in the unknown environment, the CNN model with transfer learning was designed by Park et al. [22].

As an efficient modeling technique, the CNN has shown its effectiveness in the NLOS recognition issue but some improvement measures are still expected due to its limitations. On the one hand, although the CNN model has a strong power to extract the features directly from the original signals, the measurement noises may still degrade the model's performance. Therefore, some data preprocessing measures should be considered. As studied in the present papers [23], [24], wavelet analysis can extract the time–frequency information and suppress the measurement noises. On the other, the CNN classifier only analyzes the spatial relationship of the CIR signals but does not mine the temporal correlation information sufficiently. Gramian angular field (GAF) [25], [26] is a method of transforming a time series into an image describing the temporal correlation between each time point. It provides an effective tool for enhancing the CNN model.

Based on the above discussions, this article proposes an enhanced CNN model, referred to as wavelet Gramian CNN (WGCNN), for UWB NLOS recognition by combining the wavelet analysis and the GAF. In the proposed method, the wavelet analysis is applied to extract useful trend information by suppressing the noises, and the GAF is utilized to mine the temporal correlation information of the CIR signals. Finally, a benchmark dataset will be used to verify the effectiveness of the proposed method. Generally, the contribution of this article consists of three aspects.

1) This article first introduces the GAF to convert the original CIR signals to 2-D images. To our best knowledge, although GAF has been analyzed in some other fields such as the bearing fault diagnosis [27], no existing paper has discussed the application of GAF in the UWB NLOS recognition field.
2) A three-stage UWB NLOS recognition framework is constructed including the signal preprocessing, signal transform, and the CNN classifier development. Compared with the traditional CNN model, the proposed method inserts two signal-processing stages to strengthen the CNN classification performance.
3) Different from the traditional GAF-based methods using the whole GAF image as the model input, a GAF reorganization strategy is designed to extract the local key subimages. This strategy abandons the image parts lowly correlated with the NLOS recognition and reorganizes the key subimages as the CNN inputs. They will capture the key information in GAF images for better NLOS recognition results.

II. Preliminaries

A. UWB Signal

UWB signal is commonly used in high-precision indoor positioning because of its low power consumption and high transmission rate [28], [29]. Its pulse signal width is usually less than one nanosecond, and its bandwidth usually exceeds several hundred megahertz and can reach up to several thousand megahertz. UWB was first proposed for military use. Now, it has been widely applied to many nonmilitary fields including factories, nursing homes, airports, and so on.

For most of the existing UWB-based positioning methods [30], ranging between nodes is a critical step. However, due to the possible obstacles in the indoor environments, NLOS situations affect the ranging accuracy significantly [31]. As shown in Fig. 1, the base station 2 (BS2) receives an LOS signal so that the distance between BS2 and the tag is correctly computed while the base station 1 (BS1) receives the NLOS signals, resulting in ranging errors and ultimately degrading the positioning accuracy. So, NLOS recognition is very important for indoor positioning.

By Fig. 1, BS1 and BS2 receive different signals originating from the same tag, and these signals can be viewed as the response results from different channels with the same impulse input. This procedure is called channel impulse response (CIR) [32]. In indoor positioning scenarios, CIR signals reflect the information of the channels they pass through. Therefore,

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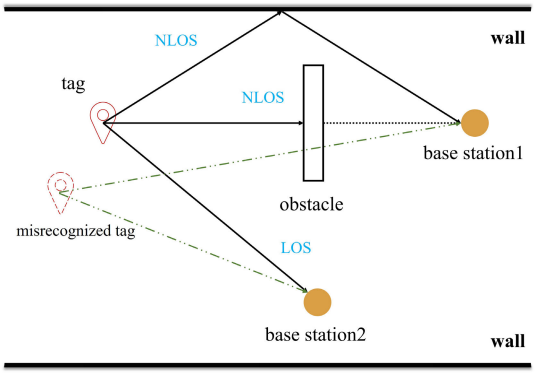


Fig. 1. LOS and NLOS signals in the UWB positioning scenario.

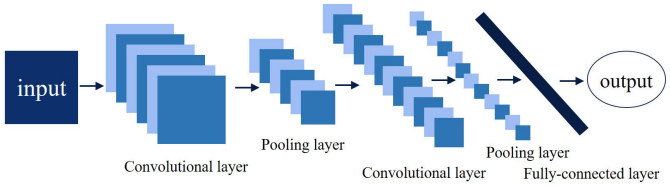


Fig. 2. CNN structure.

LOS and NLOS signals can be distinguished by investigating the CIR signals.

B. Convolutional Neural Networks

Inspired by the neural mechanism of animals, an artificial neural network (ANN) is designed to simulate the thinking behavior of animals [33], which is intrinsically a mathematical model used to handle the classification or regression tasks. Many different kinds of ANNs have been developed including back-propagation (BP) neural networks, CNNs, recurrent neural networks (RNNs), and so on. Among these models, the CNN is more suitable for image recognition task and have obtained extensive applications [34]. A typical CNN structure is shown in Fig. 2, which consists of the convolutional layer, pooling layer, and fully connected layer. The convolution layer and the pooling layer are alternately linked to extract the key features of input signals, and the extracted features are input to the fully connected layer for classification.

The convolutional layer is the most important CNN unit, which performs the convolution operation on the input signal to extract the spatial features. The convolutional procedure can be described by the following mathematical expression as [34]:

$$z(u, v) = \sum_{i=-\infty}^{\infty} \sum_{j=-\infty}^{\infty} x_{i,j} \cdot k_{u-i,v-j} \tag{1}$$

where k denotes the convolution kernel. The convolution kernel is in fact a convolutional weight matrix. In real applications, to reduce the model complexity, the weight-sharing strategy is generally adopted, which is beneficial to fast network training.

The pooling layer, also known as the downsampling layer, and its common operations are the maximal pooling and the average pooling. The pooling layer is to achieve the secondary feature of extracting and filtering. Compared with the convolution layer, the pooling layer does not contain weights. The fully connected layer is the most classic network unit which is to classify the features from the convolutional layer and pooling layer. To solve the nonlinear problem, the nonlinear activation function is usually added after each neuron to supplement the nonlinear mapping. The commonly used activation function is the rectified linear unit (ReLU) function [34]. In the training of neural networks, overfitting often affects the accuracy of network training. Therefore, to prevent overfitting of network training, a dropout layer is generally added to the full connection layer to prevent overfitting. The principle of the dropout layer is to randomly select a certain proportion of neurons to fail during each training, that is, the output value of the neurons is 0, thus reducing the degree of network overfitting.

III. Wavelet Gramian Convolutional Neural Networks

A. Model Framework

The proposed WGCNN has its framework shown in Fig. 3, which includes three steps: signal denoising, signal transformation, and classifier construction. First of all, the CIR signal is decomposed into a high-frequency part and a low-frequency part with a discrete wavelet analysis technique. The low-frequency part is retained for the following processing since it contains more valuable trend information, while the high-frequency part is discarded because it describes the measurement noises. In the second step, the 1-D low-frequency signal is then transformed into the 2-D GAF images, including the Gramian angular summation field (GASF) and the Gramian angular difference field (GADF). To reduce the image dimension and further capture the key subimages, a GAF reorganization operation is designed by cutting off the unimportant information. In the last step, the reorganized GAF images are used to train a CNN classifier for LOS/NLOS classification.

B. Signal Denoising by Wavelet Analysis

Considering that the original CIR signals may be contaminated by the measurement noises in the collection procedure, it is necessary to preprocess the signals first for strengthening the effective trend information. Here, the discrete wavelet transform (DWT) is adopted to achieve this goal. Wavelet transform (WT) [35], [36] is a signal analysis method developed on Fourier transform. The traditional Fourier transform only turns the time-domain signal into its frequency-domain expression. By contrast, WT obtains the time–frequency-domain description of the given signal. Essentially, WT is to decompose the signal into a series of wavelets, which are obtained by a basic wavelet through time shift or expansion transform. The basic formula of a continuous WT (CWT) is shown as [36]

$$\mathrm{WT}_{a,\tau} = \frac{1}{\sqrt{a}} \int x(t)\, \varphi\left(\frac{t-\tau}{a}\right) dt \tag{2}$$

where $x(t)$ is the original signal that will undergo WT, $\varphi(t)$ represents the basic wavelet, τ is the translation factor indicating the shift direction, and a is the scale factor representing

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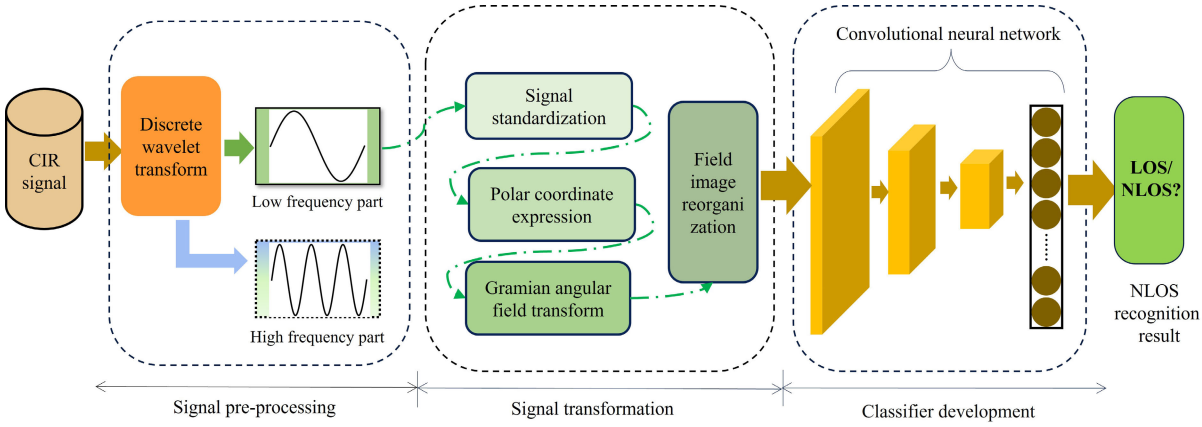


Fig. 3. WGCNN model framework.

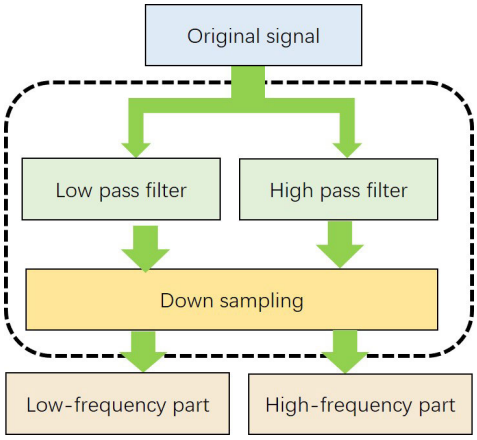


Fig. 4. Signal decomposition using WT.

the transform scale of the wavelet. Since a and τ vary continuously, it is called CWT. Because the WT has a high frequency resolution in the low-frequency part and a high time resolution in the high-frequency part, it is often used to extract local features of signals and to analyze nonstationary signals.

The DWT is the discretization form of CWT. The Mallat algorithm is a method to perform DWT easily and quickly [35]. It starts from the multiresolution decomposition of function space and realizes the fast iteration of DWT by analyzing the structure of decomposed subspace. Its transformation method is stepwise iterative decomposition, which decomposes the signal into parts at different frequencies. The decomposition framework is shown in Fig. 4.

Since there is a step downsampling operation after the filter, the length of the decomposed high-frequency part and the low-frequency part is half of the original signal. In this article, the Mallat algorithm is used to extract and downsample the CIR signal features, which is beneficial to improving training accuracy and efficiency.

C. Signal Transformation With Gramian Angular Field

The GAF [37] is a way of converting the 1-D signal to a 2-D image. A key concept of GAF is the Gramian matrix [38], which reflects the correlation of two vectors through their inner product. If both vectors are unity, then the correlation between the two vectors depends on the cosine value between the two vectors. According to this characteristic, the Gram angle field is convenient to use the cosine value to reflect the correlation between two physical quantities.

To obtain the GAF of one given signal, the first step is to scale the signal $\{x_1, x_2, x_3, x_4, \ldots, x_n\}$ to the interval $[-1,1]$ by

$$\bar{x}_i = \frac{(x_i - x_{\max}) + (x_i - x_{\min})}{x_{\max} - x_{\min}}. \tag{3}$$

Then the scaled signal is converted to the polar coordinate expression as

$$\begin{cases} \theta_i = \arccos(\bar{x}_i) \\ r_i = \dfrac{i}{N} \end{cases} \tag{4}$$

where i is the time stamp, N is the signal length, θ_i represents the angle value corresponding to the sample x_i, and the radius r_i retains the time relationship of the original signal in the polar coordinates. With (4), each signal sequence generates a unique polar coordinate map. The time information of the CIR sequence can be maintained by the radius r_i, and θ_i retains the amplitude characteristic of the original CIR signal.

Furthermore, two different GAFs, the GASF and the GADF, are obtained by [37]

$$\text{GASF} = \begin{bmatrix} \cos(\theta_1 + \theta_1) & \cdots & \cos(\theta_1 + \theta_n) \\ \vdots & \ddots & \vdots \\ \cos(\theta_n + \theta_1) & \cdots & \cos(\theta_n + \theta_n) \end{bmatrix} \tag{5}$$

$$\text{GADF} = \begin{bmatrix} \sin(\theta_1 - \theta_1) & \cdots & \sin(\theta_1 - \theta_n) \\ \vdots & \ddots & \vdots \\ \sin(\theta_n - \theta_1) & \cdots & \sin(\theta_n - \theta_n) \end{bmatrix}. \tag{6}$$

Through (5) and (6), the GASF matrix reflects the correlation between the angle sum and the GADF reflects the correlation between the angle difference. The correlation between the sum and the difference can be used to jointly express the correlation between the signal values at different time instants.

With the use of GAF, the 1-D time signal is transformed into a 2-D image, which is convenient for the CNN to train.

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To investigate (5), it is easily seen that the diagonal values of the GASF matrix are actually the original signal sequence of the signal. That means that the original data information is totally stored in the GASF diagonal line. Except for the GASF diagonal values, the other GAF elements stored the extra temporal information generated by the GASF and GADF computations. As more information is stored, the 2-D image usually has a large dimension, which may affect the efficiency of network training and may degrade the recognition accuracy. In fact, the obtained 2-D image has a sparse property, which means it contains many nonimportant elements.

To capture the key information in the GAF image, this article proposes a GAF reorganization method. Its main idea is to reorganize the GAF images with only retaining the important information for effective model training. According to the principle of GAF, its diagonal values are actually the original signal sequence. The values around the diagonal line carry important information since they reflect the correlation between two points close in time. On the contrary, the GAF values far away from the diagonal line are less important because they describe the correlation degree of two points with a long time interval. Therefore, we cut the subimages around the diagonal line and reorganize them together as the input of CNN. The reorganization operation can be described as

$$\text{Re_GASF} = \begin{bmatrix} \text{sub}_1\text{_GASF} \\ \text{sub}_2\text{_GASF} \\ \vdots \\ \text{sub}_M\text{_GASF} \end{bmatrix} \tag{7}$$

$$\text{Re_GADF} = \begin{bmatrix} \text{sub}_1\text{_GADF} \\ \text{sub}_2\text{_GADF} \\ \vdots \\ \text{sub}_M\text{_GADF} \end{bmatrix} \tag{8}$$

where $\text{sub}_m\text{_GASF}(1 \le m \le M)$ and $\text{sub}_m\text{_GADF}(1 \le m \le M)$ are the ith GASF subimage and the ith GADF subimage, respectively, and M is the number of reorganization GAF subimages.

Denote the symbols $\text{gasf}_{(i,j)}$ and $\text{gadf}_{(i,j)}$ as the (i, j)th elements of GASF and GADF, respectively, then the GASF and GADF subimages are expressed by

$$\text{sub}_m\text{_GASF} = \begin{bmatrix} \text{gasf}_{\left(1+\frac{m-1}{q},1+\frac{m-1}{q}\right)} & \cdots & \text{gasf}_{\left(1+\frac{m-1}{q},s\right)} \\ \vdots & \vdots & \vdots \\ \text{gasf}_{\left(s,1+\frac{m-1}{q}\right)} & \cdots & \text{gasf}_{(s,s)} \end{bmatrix} \tag{9}$$

$$\text{sub}_m\text{_GADF} = \begin{bmatrix} gadf_{\left(1+\frac{m-1}{q},1+\frac{m-1}{q}\right)} & \cdots & \text{gadf}_{\left(1+\frac{m-1}{q},s\right)} \\ \vdots & \vdots & \vdots \\ \text{gadf}_{\left(s,1+\frac{m-1}{q}\right)} & \cdots & \text{gadf}_{(s,s)} \end{bmatrix} \tag{10}$$

where q is the subimage dimension expressed by

$$q = \begin{cases} \dfrac{M+1}{N}, & \text{if } M \text{ is odd number} \\ \dfrac{M}{N}, & \text{if } M \text{ is even number} \end{cases} \tag{11}$$

and s is the boundary coordinate value computed by

$$s = \begin{cases} \dfrac{m+1}{q}, & \text{if } M \text{ is odd number} \\ \dfrac{m}{q}, & \text{if } M \text{ is even number.} \end{cases} \tag{12}$$

The GAF reorganization retains the data near the diagonal line of the angular field and discards the data far away from the diagonal line. This operation helps abandon the redundant information in the original GAF images and strengthens the influence of important data along the diagonal line of GAF. The best value of the reorganization subimage number M is difficult to determine by the theoretical analysis. Usually, we can select a reasonable M value by the experimental testing.

IV. Experimental Studies

A. Dataset Description

To evaluate the proposed NLOS recognition method, an open benchmark UWB dataset is employed which comes from the European Union Horizon 2020 research program [15], [19], [39]. This dataset involves seven different indoor scenarios, including office, apartment, kitchen, bedroom, and so on. In this experiment, a total of 6000 samples, including 3000 NLOS samples and 3000 LOS samples, are selected for method evaluation. Among them, 5000 samples are used to construct the model training set, while the other 1000 samples are used for method testing. Each CIR signal in the dataset is a time-series vector with 1016 samples.

The NLOS recognition performance is assessed by the classification accuracy, which is defined as

$$\text{Accuracy} = \frac{N_{\text{correct}}}{N_{\text{all}}} \times 100\% \tag{13}$$

where N_{all} is the number of testing samples and N_{correct} is the correctly recognized samples.

B. Recognition Results With WGCNN

An example of a CIR signal is illustrated in Fig. 5(a), where it is seen that both LOS and NLOS samples have their main representation interval as [701, 900]. Therefore, when we build the NLOS recognition model, only the valid data interval is selected. That means that the sample length of CIR signals is reduced from 1016 to 200. The retained signal is plotted in Fig. 5(b). Then the DWT is applied to obtain the high-frequency part and the low-frequency part as shown in Fig. 6(a) and (b). It can be intuitively seen that the low-frequency parts of LOS and NLOS have a significant difference, while the two signals in the high-frequency part are relatively similar. Furthermore, we calculate the standard variances for the high-frequency part and the low-frequency part, respectively. The results are tabulated in Table I. The high-frequency part has a standard variance of 357.059, which

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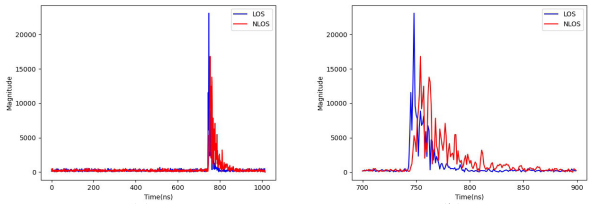


Fig. 5. LOS and NLOS signals. (a) Original signals. (b) Retained valid signals.

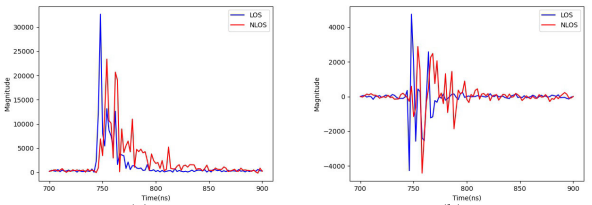


Fig. 6. Signal decomposition using DWT. (a) Low-frequency part. (b) High-frequency part.

TABLE I
PERFORMANCE COMPARISON OF THE CNN CLASSIFIERS WITH HIGH-FREQUENCY PART AND LOW-FREQUENCY PART

Model input	Standard variance	*Accuracy*
Low-frequency signal	357.059	79.64%
High-frequency signal	111.374	69.98%

is higher than that of the low-frequency part (111.374). If we use the high-frequency part and the low-frequency part as the input of the CNN classifier, respectively, the average classification accuracy is 69.98% and 79.64%, respectively, shown in Table I. Therefore, the high-frequency part can provide more valuable classification information.

The decomposed low-frequency part of the decomposed signal is scaled to the interval $[-1,1]$ according to (2) and then the scaled signal is converted into the polar coordinate representation according to (3). With the obtained polar coordinate representation, two GAFs, GASF and GADF, are computed by (4) and (5) and the corresponding images are plotted in Fig. 7. By Fig. 7, the GAF images of the LOS signal are more compact than those of the NLOS signal. This kind of difference is helpful in the LOS/NLOS classification.

GASF and GADF images represent the correlations between the angle sum and the angle difference. The single one is not enough to mine the data relationships. Therefore, the combination of the two kinds of images can better represent the temporal correlation for LOS/NLOS classification. To verify this, we construct the CNN classifiers with GASF, GADF, and GASF+GADF, respectively, and the classification results in Table II shows that the combination of two images can bring an average classification accuracy of 81.10%, which is higher than those obtained by the single GASF or GADF.

By Fig. 7, it is also observed that the GAF images are very regularly distributed. If the whole image is input to the CNN model, a lot of redundant information is involved resulting in high computation burden. Therefore, the GAF image reorganization operation is used with the procedure

TABLE II
PERFORMANCE COMPARISON OF THE CNN CLASSIFIERS WITH GASF, GADF, AND GASF+GADF

Image type	Accuracy(Avg. $\pm$ Std.)(%)
GASF	80.68 $\pm$ 0.78
GADF	79.84 $\pm$ 1.03
GASF + GADF	81.10 $\pm$ 1.28

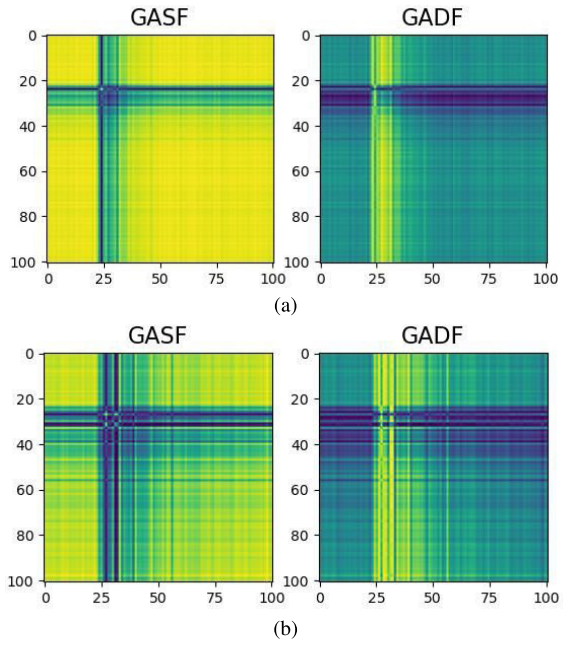


Fig. 7. GAF images corresponding to LOS and NLOS signals. (a) GASF (left) and GADF (right) images of the LOS signal. (b) GASF (left) and GADF (right) images of NLOS signal.

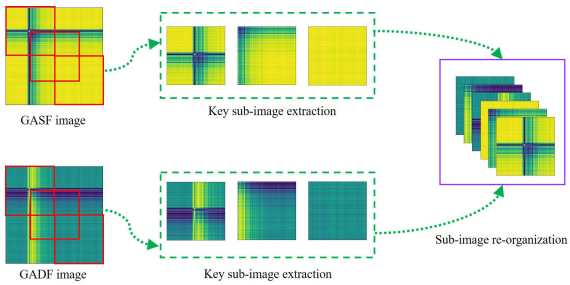


Fig. 8. Reorganization procedure of GAF images.

shown in Fig. 8. By extracting a lot of key subimages along the diagonal line, the important image parts are organized together to build the input for the CNN model.

In the reorganization procedure, a key parameter is the subimage number M. If M is set to a small number, the image reorganization role is weakened. By contrast, a too large M value may make the model focus on the local images and lead to serious information loss. By setting different values of 1, 2, 3, 4, 5, 7, and 9, the classification accuracy of the proposed method is tabulated in Table III and plotted in Fig. 9. In Table III, Avg. represents the average accuracy value and Std. denotes the standard variance of accuracy. The results show that the WGCNN model with $M = 4$ has the best classification performance, which has an average accuracy of 82.63% with a standard variance of 0.89%.

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TABLE III
RECOGNITION ACCURACY VALUES OF WGCNN MODELS WITH DIFFERENT REORGANIZATION NUMBER *M* SETTINGS

re-organization number	Avg. ± Std. (%)
$M = 1$ (no re-organization)	81.38 ± 0.78
$M = 2$	81.54 ± 0.54
$M = 3$	82.36 ± 0.99
$M = 4$	82.63 ± 0.89
$M = 5$	80.79 ± 0.73
$M = 7$	81.51 ± 0.87
$M = 9$	81.10 ± 1.14

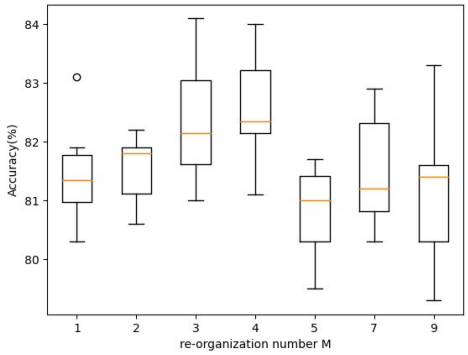


Fig. 9. WGCNN model recognition performance.

TABLE IV
CNN STRUCTURE PARAMETERS USED IN THE WGCNN MODEL

Network Structure	Parameter	Value
1st convolutional layer	n_{ic}, n_{oc}, n_{ck} [1]	8, 20, 4
1st average pooling layer	n_{pk}, n_{str} [2]	2, 2
2nd convolutional layer	n_{ic}, n_{oc}, n_{ck}	20, 60, 4
2nd average pooling layer	n_{pk}, n_{str}	2, 2
1st fully-connected layer	n_{in}, n_{out}, r_{drop} [3]	960, 500, 0.3
2nd fully-connected layer	n_{in}, n_{out}, r_{drop}	500, 2, 0

Note:
[1] n_{ic}, n_{oc}, and n_{ck} represent the input channel number, the output channel number, and the convolutional kernel size, respectively.
[2] n_{pk}, n_{str} denote the pooling kernel size, and the stride size, respectively.
[3] n_{in}, n_{out}, and r_{drop} represent the input number, the output number and the dropout rate of fully-connected layer, respectively.

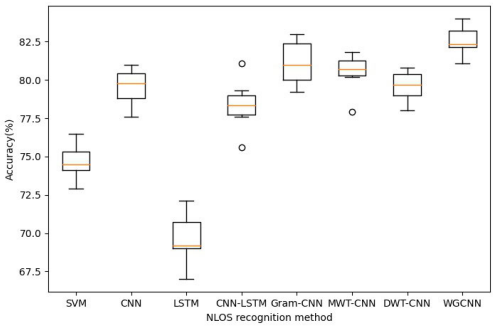


Fig. 10. Comparison of different NLOS recognition models.

C. Performance Comparison of Different Methods

To further verify the effectiveness of the process NLOS recognition method, several representative deep-learning-based methods are utilized for method performance comparison. The compared methods include as follows.

1) *SVM:* This is the traditional machine-learning model, which applies SVM to handle the original CIR signals directly.
2) *CNN:* This is the basic model, which applies the CNN to handle the original CIR signals directly [20].
3) *LSTM:* It is also a famous deep neural network model, often used in time series analysis.
4) *CNN-LSTM:* It combines the LSTM and CNN to build the LOS/NLOS classifier [19].
5) *Gram-CNN:* In this model, the original CIR signals are first transformed into 2-D images, then the CNN model is built for classification.
6) *MWT-CNN:* This method applies Morlet wavelet analysis to handle the CIR signals and then uses CNN to construct the recognition model [23].
7) *DWT-CNN:* This method applies DWT to obtain the trend information of CIR signals and then uses CNN to recognize the NLOS signals.
8) *WGCNN:* It is the method proposed in this work. By the previous analysis related to Table III, the reorganization number M is set to 4.

All the compared methods have the same CNN models except for the MWT-CNN. The MWT-CNN structure is from [23]. The other used CNN models are preassigned with two convolution layers and two average pooling layers. The CNN structure parameters used in WGCNN are listed in Table IV. The input of the CNN is the reorganized GAF subimages. Its size is determined by the size of subimages. In the case of $M = 4$, the dimension of subimage is 25×25. Further considering the input channel number $n_{\text{ic}} = 8$ at the first convolutional layer, the input size of the CNN model is $8 \times 25 \times 25$. The output of the CNN indicates the classification result is LOS or NLOS. Since the CNN output adopts the one-hot encoding, the output size is 1×2. In addition, the model training epoch is set to 300 times, the batch size is set to 64, and the learning rate is set to 0.00005.

All seven models were trained ten times, respectively, to avoid the possible random influence. The recognition results are shown in Table IV and Fig. 10. In Table IV, Avg. represents the average accuracy value, Std. denotes the standard variance of accuracy, Max. is the maximal accuracy value, while Min. is the minimal accuracy value among ten modeling trials.

From Table IV, the average accuracies of SVM and CNN are 74.74% and 79.57%, respectively. With the use of wavelet analysis, MWT-CNN and DWT-CNN achieve a little higher performance with accuracy values of 80.57% and 79.64%, respectively. If the GAF is introduced, the Gram-CNN can obtain an average accuracy of 81.10%. So, the GAF technique provides more attribution to performance improvement than the wavelet analysis. To combine the two techniques of wavelet analysis and GAF, the proposed WGCNN method has the highest accuracy 82.63%. Moreover, among the ten training trials, the WGCNN method has the highest maximal and minimal accuracy at 84.0% and 81.1%. It is noted that the LSTM model seems not to benefit the classification performance. With the use of LSTM, the CNN-LSTM method's average accuracy even degrades to 78.37%. To sum up, the proposed

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TABLE V
NLOS RECOGNITION ACCURACY VALUES OF DIFFERENT CNN-BASED METHODS ON THE BENCHMARK DATASET

Method	Avg. ± Std.(%)	Max.(%)	Min.(%)	Training time(s)	Testing time(s) per sample
SVM	74.74 ± 1.04	76.50	72.90	2.58	7.96×10^{-4}
CNN	79.56 ± 1.05	81.00	77.60	9.95	8.01×10^{-6}
LSTM	69.62 ± 1.49	72.10	67.00	337.43	5.90×10^{-5}
CNN-LSTM	78.37 ± 1.34	81.10	75.60	126.19	1.81×10^{-5}
Gram-CNN	81.10 ± 1.28	83.00	79.20	270.94	1.68×10^{-4}
MWT-CNN	80.57 ± 1.03	81.80	77.90	575.17	1.71×10^{-4}
DWT-CNN	79.64 ± 0.78	80.80	78.00	32.88	7.08×10^{-6}
WGCNN	82.63 ± 0.89	84.00	81.10	131.61	2.41×10^{-5}

method demonstrated its validity in the NLOS recognition scene.

Compared with the basic CNN model, the proposed WGCNN method is more complicated with a three-level structure. Although the model becomes complicated, the increased computation cost mainly lies in the model training stage. This stage is usually performed before the UWB positioning system is constructed. Once the model training is finished, the testing complexity on the new sample is very low. The model complexity can be reflected by the model training time and the testing time, which are listed in Table V for all the used methods. By Table V, it is seen that the CNN model training time is 9.95 s, while the WGCNN model training time is 131.61 s. This verifies WGCNN is more complicated than CNN. The testing times per sample for CNN and WGCNN are 8.01×10^{-6} and 2.41×10^{-5}, respectively. Both testing times are very small. Therefore, the difference in the testing complexity can almost be omitted and the proposed method can be helpful to the improvement of UWB portioning performance.

V. CONCLUSION

In this article, we propose a modified CNN model WGCNN for better UWB NLOS recognition. This proposed method applies the DWT to extract the main trend of CIR signals and further integrates the GAF technique to transform the original 1-D signal into 2-D GADF and GASF images. This method not only utilizes the spatial data information, but also makes good use of the temporal data information. Moreover, the GAF reorganization is designed to highlight the key subimages for strengthening the model performance. The application results on the benchmark dataset show that the proposed WGCNN method increases the average recognition accuracy by 3.09% compared with the basic CNN model. The maximal and minimal accuracy of ten modeling trials obtain increments of 3% and 3.5%, respectively.

Although this article makes some progress in CNN-based NLOS recognition, some related research problems should not be omitted. One is the data imbalance issue. In most cases, the number of NLOS signals may be much smaller than the NLOS signal number. The balanced classification is an ideal case. How to build an effective model with a small quantity of NLOS data is a valuable problem. Another issue is about how to take advantage of prior knowledge. Although the deep-learning model avoids dependence on manual feature engineering, effective feature construction using prior knowledge will be very helpful to enhance the deep-learning model. All these will be studied in future work.

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